Health Tech Regulatory & Compliance Analyst Roadmap

Action Kit

Navigate FDA SaMD regulations, HIPAA compliance, AI governance frameworks, and quality management systems to ensure health tech products meet regulatory requirements and reach market safely.

Difficulty	Moderate
Timeline	4 to 8 months
Last Updated	2026-03-24

This action kit contains your 90-day execution plan, portfolio project templates, resume bullet formulas, interview preparation questions, and resource checklist.

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Section 1: 90-Day Execution Plan

This plan maps directly to your learning path phases. Each month has specific deliverables and checkpoints to keep you on track.

Month 1: Regulatory Foundations

Hours: 80 to 120

- FDA Medical Device Regulatory Framework (510(k), De Novo, PMA pathways)
- Software as a Medical Device (SaMD) classification and regulation
- HIPAA Privacy and Security Rule deep dive
- Quality Management System fundamentals (ISO 13485)
- Regulatory Affairs terminology and submission types

Checkpoint: Map the regulatory pathway for a hypothetical AI-powered clinical decision support tool. Determine whether it qualifies as a medical device under FDA guidance, identify the appropriate submission pathway, and document the key regulatory requirements.

Month 2: Compliance Systems and Standards

Hours: 100 to 140

- Quality Management Systems (QMS) implementation and auditing
- Risk Management for Medical Devices (ISO 14971)
- Software Lifecycle Processes (IEC 62304)
- AI/ML Regulatory Frameworks (FDA PCCP guidance, EU AI Act basics)
- Design Controls and Design History File management
- Cybersecurity requirements for health tech (FDA premarket cybersecurity guidance)

Checkpoint: Complete a mock regulatory submission package for an AI-enabled health tech product. Include risk analysis (ISO 14971), software documentation (IEC 62304), design control records, and a cybersecurity plan. Present the package as a portfolio deliverable.

Month 3: Specialization and Practice

Hours: 60 to 100

- Track A: SaMD and AI/ML Device Regulation (FDA PCCP, predetermined change control, continuous learning systems)
- Track B: Healthcare Privacy and Data Governance (HIPAA compliance programs, state privacy laws, international data protection)
- Track C: Clinical Investigation and Evidence (IDE submissions, clinical study regulatory requirements, real-world evidence)
- Track D: International Regulatory Strategy (EU MDR, EU AI Act, Health Canada, MHRA frameworks)

Checkpoint: Complete two specialization projects: (1) a regulatory gap analysis for an existing health tech product against current FDA guidance and (2) a compliance program design or international regulatory strategy comparison. Both become portfolio deliverables.

Section 2: Portfolio Project Templates

Each project below includes a dataset, tools, timeline, and your clinical advantage. Complete at least 2 to 3 of these for a competitive portfolio.

Project 1: FDA SaMD Regulatory Pathway Analysis

Select a real AI-enabled health tech product (or design a hypothetical one). Determine its SaMD classification using the IMDRF framework, identify the correct FDA submission pathway (510(k), De Novo, or PMA), and create a regulatory strategy document with timeline and key milestones.

Dataset	FDA AI/ML-Enabled Medical Device Database
URL	https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-lea
Tools	FDA Guidance Documents, IMDRF Framework, Regulatory Strategy Template
Time Estimate	4 to 6 weeks

Clinical Advantage: You can evaluate whether the product's intended clinical use is realistic and whether the regulatory claims align with actual clinical workflows

Project 2: HIPAA Compliance Program Design

Design a comprehensive HIPAA compliance program for a digital health startup. Include risk assessment methodology, policies and procedures, training requirements, breach notification protocols, and Business Associate Agreement templates.

Dataset	HHS Breach Portal (Wall of Shame)
URL	https://ocrportal.hhs.gov/ocr/breach/breach_report.jsf
Tools	NIST Cybersecurity Framework, HHS HIPAA Resources, Risk Assessment Templates
Time Estimate	5 to 7 weeks

Clinical Advantage: You understand which PHI touchpoints create the highest risk because you have worked with patient data at the point of care

Project 3: AI/ML Device Predetermined Change Control Plan (PCCP)

Create a PCCP for a hypothetical AI-enabled clinical decision support tool. Document the modification protocol, performance monitoring plan, update validation methodology, and transparency requirements per FDA guidance.

Dataset	FDA PCCP Guidance and authorized AI/ML devices list
URL	https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-software-medica
Tools	FDA PCCP Guidance, ISO 14971 Risk Management, IEC 62304 Software Lifecycle
Time Estimate	4 to 6 weeks

Clinical Advantage: You can assess whether proposed AI model changes could create clinical safety risks that technical teams might underestimate

Project 4: Regulatory Gap Analysis: Existing Health Tech Product

Select a commercially available health tech product and perform a regulatory gap analysis against current FDA requirements. Identify compliance gaps, prioritize risks, and propose a remediation plan with timeline and resource estimates.

Dataset	FDA 510(k) Database and MAUDE adverse event database
URL	https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfmaude/search.cfm
Tools	FDA MAUDE Database, 510(k) Database, Gap Analysis Framework
Time Estimate	3 to 5 weeks

Clinical Advantage: Your clinical experience helps you identify gaps that matter most for patient safety, not just technical compliance

Project 5: International Regulatory Strategy Comparison

For a hypothetical digital health product, compare the regulatory requirements and timelines for market entry in the US (FDA), EU (EU MDR + EU AI Act), Canada (Health Canada), and UK (MHRA). Deliver a market entry strategy recommendation.

Dataset	Regulatory agency databases and guidance documents
URL	https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance
Tools	FDA Guidance, EU MDR Text, EU AI Act, Health Canada MDR
Time Estimate	5 to 7 weeks

Clinical Advantage: Your understanding of clinical practice variations across countries adds depth to regulatory strategy that purely legal analysis misses

Section 3: Resume Bullet Formulas

Use these formulas to translate your clinical experience into language that resonates with hiring managers in health data roles. Replace bracketed text with your specifics.

Nursing Background

- Leveraged [clinical skill: Patient safety event reporting and root cause analysis] to deliver [tech outcome: Post-market surveillance and adverse event documentation (FDA MDR/MedWatch)], resulting in [quantified impact].
- Leveraged [clinical skill: Joint Commission survey preparation and accreditation compliance] to deliver [tech outcome: Quality Management System (QMS) auditing and ISO 13485 compliance], resulting in [quantified impact].
- Leveraged [clinical skill: Clinical protocol adherence and documentation standards] to deliver [tech outcome: Design control documentation and regulatory submission preparation], resulting in [quantified impact].
- Leveraged [clinical skill: HIPAA awareness in daily clinical workflows] to deliver [tech outcome: Privacy impact assessments and HIPAA compliance program management], resulting in [quantified impact].

Pharmacy Background

- Leveraged [clinical skill: DEA controlled substance compliance and state board regulations] to deliver [tech outcome: Multi-jurisdictional regulatory compliance and submission management], resulting in [quantified impact].
- Leveraged [clinical skill: Drug safety monitoring and adverse drug reaction reporting] to deliver [tech outcome: Pharmacovigilance systems and post-market safety surveillance], resulting in [quantified impact].
- Leveraged [clinical skill: Formulary review and drug monograph evaluation] to deliver [tech outcome: Regulatory submission review and clinical evidence evaluation], resulting in [quantified impact].
- Leveraged [clinical skill: Pharmacy and Therapeutics committee participation] to deliver [tech outcome: Regulatory advisory boards and standards committee participation], resulting in [quantified impact].

Other Clinical Background

- Leveraged [clinical skill: Evidence-based practice and clinical guideline evaluation] to deliver [tech outcome: Clinical evidence assessment for regulatory submissions (510(k), De Novo, PMA)], resulting in [quantified impact].
- Leveraged [clinical skill: Clinical documentation and medical record standards] to deliver [tech outcome: Technical documentation standards (IEC 62304, ISO 13485 documentation requirements)], resulting in [quantified impact].
- Leveraged [clinical skill: Quality improvement projects (PDSA cycles, Lean Six Sigma)] to deliver [tech outcome: Corrective and Preventive Action (CAPA) processes and continuous improvement], resulting in [quantified impact].
- Leveraged [clinical skill: Informed consent processes and patient rights] to deliver [tech outcome: Human subjects protection and clinical investigation ethics (IRB submissions)], resulting in [quantified impact].

Pro Tip: Always quantify. Instead of 'analyzed data,' write 'analyzed 50,000+ claims records to identify $2.3M in recoverable overpayments.'

Section 4: Interview Preparation Questions

Practice these role-specific questions. For each, prepare a 2-minute answer that highlights both your technical skills and clinical background.

Technical Questions

1. How would you determine whether a health tech product qualifies as a Software as a Medical Device (SaMD) under FDA guidelines?
2. Walk me through the key differences between FDA 510(k), De Novo, and PMA pathways.
3. How do you conduct a HIPAA risk assessment for a new health tech application?
4. Describe the ISO 14971 risk management process and how clinical experience informs hazard identification.
5. How would you assess a health tech company's compliance readiness for the EU AI Act?

Behavioral Questions

1. Tell me about a time your clinical background helped you identify a regulatory risk that others overlooked.
2. How do you handle a situation where a product team wants to ship a feature that has unresolved compliance questions?
3. Describe how you communicate complex regulatory requirements to non-technical stakeholders.
4. How do you stay current with rapidly evolving health tech regulations across multiple jurisdictions?

Scenario Questions

1. The FDA sends a warning letter about a product your company distributes. Walk me through your response plan.
2. A clinical AI model update triggers a question about whether it needs a new regulatory submission. How do you evaluate this?
3. A customer in the EU asks for your company's AI Act compliance documentation, but your company has not started preparing. What do you do?

Section 5: Resource Checklist

Use this checklist to track your progress through all recommended resources.

Certifications

Certification	Signal	Cost	Timeline
RAC (Regulatory Affairs Certification)	High	$350 to $500 exam (RAPS member discount available)	3 to 6 months study after building foundational knowledge
CHC (Certified in Healthcare Compliance)	High	$295 to $595 exam	Requires 1 year compliance experience or 1,500 hours direct compliance work plus 20 CEUs
CHPC (Certified in Healthcare Privacy Compliance)	High	$295 to $595 exam	Similar requirements to CHC with privacy focus
ISO 13485 Lead Auditor Certification	High	$1,500 to $3,000 (includes training course)	5-day training course plus exam
Certified Quality Auditor (CQA) from ASQ	Helpful	$294 to $494 exam	8 to 12 weeks study; requires 8 years work experience (education can substitute)
CIPP/US (Certified Information Privacy Professional)	Helpful	$550 exam	4 to 8 weeks study

Learning Resources by Phase

Regulatory Foundations

- [Free] FDA Digital Health Center of Excellence: https://www.fda.gov/medical-devices/digital-health-center-excellence
- [Free] FDA SaMD Guidance Documents: https://www.fda.gov/medical-devices/digital-health-center-excellence/software-medical-device-samd
- [Free] HHS HIPAA for Professionals: https://www.hhs.gov/hipaa/for-professionals/index.html
- [Free] IMDRF SaMD Framework: https://www.imdrf.org/documents/software-medical-device-samd-key-definitions
- [Free] FDA Learning Portal: https://www.fda.gov/training-and-continuing-education
- [Paid] RAPS Regulatory Fundamentals Course: https://www.raps.org/education
- [Paid] Coursera Regulatory Affairs Specialization: https://www.coursera.org

Compliance Systems and Standards

- [Free] FDA Quality System Regulation (21 CFR Part 820): https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-820

- [Free] FDA AI/ML-Based SaMD Action Plan: https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-software-medical-device
- [Free] NIST AI Risk Management Framework: https://www.nist.gov/artificial-intelligence
- [Free] FDA Cybersecurity Guidance: https://www.fda.gov/medical-devices/digital-health-center-excellence/cybersecurity
- [Free] Greenlight Guru Academy: https://www.greenlight.guru/academy
- [Paid] ASQ Certified Quality Auditor Resources: https://asq.org/cert/quality-auditor
- [Paid] Udemy ISO 13485 QMS Implementation: https://www.udemy.com

Specialization and Practice

- [Free] FDA Predetermined Change Control Plan Guidance: https://www.fda.gov/regulatory-information/search-fda-guidance-documents
- [Free] EU AI Act Official Text: https://artificialintelligenceact.eu
- [Free] HCCA Healthcare Compliance Resources: https://www.hcca-info.org
- [Free] ClinicalTrials.gov: https://clinicaltrials.gov
- [Paid] RAPS RAC Exam Preparation: https://www.raps.org/rac
- [Paid] HCCA CHC Certification Preparation: https://www.hcca-info.org/certification/become-certified/chc

Your First Three Moves

Map the regulatory landscape (3 hours)

- Read the FDA Digital Health Center of Excellence overview page
- Browse the FDA's list of AI/ML-enabled authorized medical devices
- Read 3 to 5 real 510(k) summaries for digital health products on the FDA database

Audit your own compliance knowledge (2 hours)

- List every compliance, accreditation, or quality requirement you encountered in clinical practice
- Match each one to a corresponding health tech regulatory requirement (HIPAA, ISO 13485, FDA QSR)
- Identify the 3 biggest gaps between your current knowledge and regulatory job requirements

Start learning the regulatory framework (30 minutes daily for 4 weeks)

- Complete FDA Learning Portal modules on device regulation basics
- Join RAPS (Regulatory Affairs Professionals Society) as a student or early career member
- Start following FDA Digital Health Twitter/LinkedIn for real-time regulatory updates

Ready to go deeper? Take the free career quiz at quiz.ehoneahobed.com or subscribe to The Transmutation newsletter at thetransmutation.substack.com